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Model-Authored Improvers for Battery-Free Design

Working draft

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Abstract

Can changing a method-proposal procedure improve battery-free design under a fixed scientific task and a complete experimental cost boundary? We compare ordinary reflection with one model-authored replacement using three paired descendant lineages, a shared historical packet, and fresh design episodes. All 30 effective selection cells pass. Among 27 final cells, the unchanged base method, ordinary-reflection descendants, and replacement descendants obtain 7/9, 6/9, and 8/9 complete-service PASS outcomes. All 18 cells on two digital derivative tasks pass; differences occur only on the original event service under a previously unexposed waveform. Four non-submissions end at a 32,768-token response limit and remain in denominators and costs. Full actual ordinary-reflection and replacement pipelines cost 135 and 138 requests; after removing only four retained apparatus incidents, they cost 123 and 129. The 77 actual attempts use 367 requests in total. This is a descriptive result conditional on one shared improver formation, not independent training replication or a stable advantage. Default method promotion, recursive self-improvement, and physical qualification are not established.

1 Experimental question and objects

The experiment asks whether changing the procedure that proposes a design method changes the service achieved by its descendants and the resources needed to obtain them. Three objects have distinct roles. A design D is a task-specific device or timing artifact. A method H is procedural text supplied at design-task entry. An improver G is the proposal procedure used to generate H; it is neither a device submission nor a method already in force. Writing a new G establishes an authored artifact. Establishing a useful change requires fresh design decisions, an unchanged service evaluator, and costs that include formation, proposal, selection, and execution.

The reference H_0 and ordinary-reflection procedure G_0 are inherited unchanged from phase 3. One prospective episode formed G_1. Its allowed change was confined to the natural-language proposal procedure. Tasks, physical-model semantics, default H_0, proposal prefix, source access, tools, model configuration, budgets, evaluator, and selection rule remained fixed. There was no weight update. Three descendant lineage pairs subsequently used the one frozen G_1. They are repeated descendants conditional on a shared formation, not three independent improver training events.

The registered finite experiment comprises 73 logical cells and, after an explicit cache-representation amendment, 77 actual attempts. All are closed. The reference method and the two descendant arms obtain 7/9, 6/9, and 8/9 final PASS outcomes, respectively. The numerical difference is confined to one original-service task; it is not evidence of a stable improver advantage. This report binds the protocol, amendment, original artifacts, selection freeze, complete results, and actual accounting [1, 2].

2 Information, pairing, and frozen decisions

2.1 Common historical information

Both proposal arms receive the same 75,955-byte mechanically projected model history. The complete 257,661-byte packet remains bound for host audit; it is not the packet delivered to the model. This distinction matters when describing available evidence. The histories contain earlier public experience, including failures and qualification limits, rather than final test feedback. The separate original-service development transfer, O0, is excluded from G_1 formation and all G/H proposal and selection inputs.

The configured provider reports the alias `deepseek-flash`. The actual configuration is hash-bound, but independently pinned model weights remain UNKNOWN. The claim is therefore conditional on the recorded configured provider, not a verified immutable weight checkpoint. The public tool projection is common to all arms: arrays longer than twelve expose their count, full canonical hash, first six entries, and last six entries. Host records retain the full response. Public feedback and host service evaluation remain separate.

2.2 Fresh descendants and design episodes

For lineage $\ell \in \{1, 2, 3\}$ and arm $g \in \{0, 1\}$, two named sibling methods are proposed independently:

$$H_{\ell,g,j} \leftarrow \text{Propose}(G_g, H_0, \mathcal{E}_{\text{public}}), \qquad j \in \{1, 2\}.$$

Each sibling starts from H_0 and the identical history, rather than editing its sibling's output. This gives twelve actual H proposals. A consumed descendant consists of unchanged H_0, two newlines, and the original proposed method text. Its hash therefore differs from the proposal-text hash; no host rewriting is implied. Each eligible method receives two fresh design episodes on `b-pulse-fast` and `b-volatile-reset`. Six fresh H_0 control episodes provide the same two tasks in each lineage. Thus selection comprises twenty-four descendant cells and six common-control cells.

Every D episode starts at task entry with the registered tools, information, caps, and frozen initial exact-input cache, plus an isolated overlay. A cache hit reuses a physical-model response for exactly matching input; it does not reuse an Agent decision. It still consumes the registered logical design exposure. A changed input cannot inherit the old physical response. Repeated Agent decisions that share one cached response are not independent physical experiments.

All thirty effective selection cells closed, and all six arm selections were frozen before any final input was delivered to a participant. The sealed preservation audit verifies this ordering and actual payload derivation [3, 4]. Sibling ranking uses, in order, more complete-service PASS outcomes, fewer wrong service judgments, fewer charged requests for the proposal plus its two effective selection episodes, and stable child index. If neither sibling is admissible, the protocol requires an explicitly labeled H_0 fallback. Invalid, no-update, failed, and censored attempts remain recorded. They do not authorize replacement proposals or a change to the default H_0.

The final comparison contains twenty-seven fresh D episodes: three final tasks for six selected descendants and three lineage-matched H_0 controls. These tasks cover a within-model battery-free derivative, a C-lite service derivative, and the original service under a separately frozen, previously unexposed waveform. Unexposed means absent from formation and selection; a final D participant normally receives the complete frozen current-task facts, including its waveform. This is not a hidden-waveform test of the final designer. Final outcomes were not used to reselect, tune, repair, or promote a method.

3 What the model actually formed

3.1 One frozen proposal procedure

The actual G_1 is a 4,599-character procedure. It asks future proposals to start with a source and capability ledger, express a typed symbolic method, and branch on the consuming world's latch and source characteristics. It then asks the proposer to triage recorded delivery, rehearse the procedure against retained history, state a

source-bound and tick-bound falsifier, retain errors and costs, use the proposal channel appropriate to H, and return no-update when support is absent. Historical numerical examples are included as examples; they are not a new D submission.

These are descriptions of the model's instructions, not accepted scientific rules. In particular, its blanket treatment of events after the last transmission as benign can fail for ongoing or multiple-event services. Required rehearsal, exact source fields, and mandatory UNKNOWN judgments can also constrain useful action or fail when a consuming interface differs. The procedure and its descendants are preserved as submitted; the host does not silently correct their formulas, channel assumptions, or service logic.

3.2 Closed formation evidence

One actual provider episode formed G_1 with three requests. Twelve actual provider episodes then submitted twelve proposed methods, using forty-three requests. Raw and normalized submissions are retained. Host readback binds each descendant to H_0, its assigned G, the common model history, and its submitted method hash. All twelve received artifact-scope semantic admission: they change a procedural H rather than submitting a device or modifying the apparatus. Every audit explicitly leaves scientific validity `NOT_ESTABLISHED`. Admission does not certify the proposed arithmetic, rehearsal claims, transfer, or eventual service performance.

Component	Episodes	Requests	Upper USD
Shared G_1 formation	1	3	0.0269157
G_0 descendants	6	20	0.1832424
G_1 descendants	6	23	0.2658975
Formation and proposals	13	46	0.4760556

Table 1: Actual formation and proposal costs only; selection, final execution, controls and host incidents are additional.

The G_0 descendants contain 6,215–9,014 Unicode characters of method text; the G_1 descendants contain 9,300–11,054. Longer text is an observed artifact property in this sample, not evidence of a better method. Summed episode wall time is 38.723294 seconds for formation, 352.501470 for the six G_0 proposals, and 510.444238 for the six G_1 proposals. These sums are not elapsed campaign time.

Gateway logs contain fifteen `UNKNOWN_PUBLIC_SOURCE` failures across five H episodes, three in each. The failed acquisitions and their costs remain in the original episodes. The thirteen formation/proposal ledgers have no unknown or missing provider usage. A tool reporting an unavailable source is distinct from UNKNOWN billing usage. Participant accounts of schema rejection or native calls are not substituted for trusted logs: the retained audit notes an unestablished schema-rejection claim and a conflation of source acquisition with native engineering dispatch. Formation/proposal host records report zero native dispatches.

4 Apparatus incident and registered recovery

An explicitly retained hold snapshot records three selection episodes with `KeyError: 'trace'` and a fourth censored episode. The historical cache loader supplied `observations.json` without the separately sealed `trace.csv` required by assessment. This is a representation failure in the apparatus; it cannot establish inferiority of either improver arm. At that snapshot, nine selection attempts had closed, five were valid, and no final episode had closed. These are historical snapshot counts, not a claim about the current campaign state.

The approved amendment changes historical CSV loading and provenance checks [5]. It preserves original v1 records, compressed raw responses, task semantics, methods, participant payloads, model, service, and the original tighter request and wall-time ceilings. Six existing-data qualification checks passed: one restored historical assessment, invariance of all five valid selections, and rejection of missing trajectory, wrong trajectory

binding, wrong exact input, and wrong world. Those recorded checks used no new provider request or simulation; they are not hardware qualification.

Four prospectively named `recovery-v11` identities then ran with fresh contexts and the original payload and initial cache. Their sealed receipt records four actual complete-service PASS outcomes, twenty provider requests, and an upper cost of USD 0.1182666. This establishes the reported recovery outcomes only. The four original faulty/censored attempts remain separately visible: twenty-one requests and USD 0.1318473 of host-incident overhead. They are not overwritten, relabeled as method failures, or erased from arm and campaign cost. Neither G_1 nor any H was regenerated. The amendment retains seventy-three canonical logical slots while permitting at most seventy-seven actual attempts, including the four original attempts.

5 Selection cost and full pipeline cost

Let $c(e)$ denote the recorded resource vector for episode e: charged requests, summed episode wall time, and settled ledger upper USD. Formal sibling ranking uses the proposal and its two effective selection episodes. Under the amendment, each affected canonical selection slot resolves to its named recovery identity; original incident attempts are separately itemized. This formal ranking cost must not be mistaken for the cost of obtaining the arm's result. Full arm accounting requires

$$C_{\ell,g} = A_{\ell,g} + \sum_{j=1}^{2}\left[c(P_{\ell,g,j}) + \sum_{s=1}^{2} c(S^{\mathrm{eff}}_{\ell,g,j,s})\right] + \sum_{t=1}^{3} c(F_{\ell,g,t}) + c(I_{\ell,g}),$$

where A is allocated formation cost, P a proposal, S^{eff} an effective selection episode, F a final episode under the frozen selected method or fallback, and I the retained incident attempts. Both siblings, including the unselected sibling, are charged. Common H_0 controls and the separate initial transfer experiment are accounted once at campaign level, not copied into every arm or omitted from a reported campaign total.

The single G_1 event is counted once globally. Its three requests allocate one to each G_1 lineage; USD 0.0269157 allocates USD 0.0089719 each. Wall time is divided in conserved decimal thirds, with the last allocation absorbing the division residue. These allocations do not create additional formation events. Each arm is capped at 120 requests, 6,600 summed wall seconds, and USD 10, including allocated formation and failures. An H proposal has eight requests and 400 seconds; a D episode has twelve requests, 800 seconds, forty tool calls, and two distinct logical designs. Global ceilings remain 860 requests, 54,400 summed wall seconds, USD 100, and 150 new scientific executions. Estimates and ledger upper bounds remain distinct from invoices; human and orchestration assistance remains nonzero and unpriced.

6 Complete results

6.1 Separate original-service development transfer

The registered O0 development experiment contains a candidate and fixed periodic and rich-state controls [6]. The candidate and rich-state control pass nominal complete event service, while the periodic control fails. The candidate's own judgment is UNKNOWN despite host PASS (Table 2). Matching the rich-state control establishes no benefit beyond it. These three episodes use eighteen requests and USD 0.1579020 upper; their results are excluded from the common G/H history. Original service contract, physical UNKNOWN, and earlier A/R failures are retained.

6.2 Selection and final service

All 30 effective selection cells pass complete service with zero nominal judgment disagreements. Thus sibling ranking reduces to proposal-plus-two- selection request count, then stable child index. Lineages 2 and 3 under G_0 tie on requests and select child 1; USD is not a tie breaker. The other four arms select child 2 (Table 3). All twelve siblings remain eligible; none is discarded or repaired for predicted final performance. The six selected composite method hashes are frozen before final exposure [3, 4].

Condition	Host service	Participant	Requests	Upper USD
candidate	PASS	UNKNOWN	8	0.0760389
fixed-periodic	FAIL	FAIL	5	0.0321168
rich-state	PASS	PASS	5	0.0497463

Table 2: The separate O0 development transfer. UNKNOWN disagrees with the candidate's host PASS under the nominal judgment metric. O0 contributes no history to G/H formation or selection.

Arm	$q(h_1)$	$q(h_2)$	Selected	Consumed H hash
L1-G_0	14	13	2	`76060e3a5bb3`
L1-G_1	15	12	2	`ba3c85390ea1`
L2-G_0	13	13	1	`bc7d9af16574`
L2-G_1	14	13	2	`206485c17bc5`
L3-G_0	15	15	1	`dde0e5465cf7`
L3-G_1	14	13	2	`1d206feb1c02`

Table 3: All twelve siblings pass both selection tasks, with zero wrong service judgments; all six H_0 controls also pass (30/30 effective cells). Requests include each child's proposal and two effective selections. Tied request counts use stable child index, not USD. Consumed H is unchanged H_0 followed by the original proposed text; hashes refer to this composite.

The 27 final cells produce 21 PASS, two submitted FAIL, and four incomplete non-submissions (Table 4). The battery-free stretched-window task and C-lite half-power query task each pass 9/9, totaling 18/18 across arms. Only the original event service under its unexposed development/selection waveform separates outcomes: H_0 yields one PASS, one submitted FAIL and one censored cell; G_0 yields zero, one and two; G_1 yields two, zero and one. Appendix A lists every final cell's design parameters, requests and cost. The host's complete-service judgment remains separate from the participant's assessment [2].

Arm	Cells	PASS	Submitted FAIL	Censored	Disagreements
H_0	9	7	1	1	1
G_0	9	6	1	2	1
G_1	9	8	0	1	0

Table 4: All 27 registered final cells, including four token-censored non-submissions. Disagreements compare a submitted participant judgment with the host service result; censored cells have no submitted judgment.

Both submitted O failures, L1-H_0 and L3-G_0, are judged PASS by their participants but rejected by the host for `WRONG_EVENT_TIME`. Both use one-second sampling. The two successful G_1 O designs use 0.75-second sampling, and L2-H_0 also succeeds at 0.75 seconds. The observed design choice is therefore not a mechanism unique to G_1. Common submitted O choices include 15 mF storage, nonvolatile event cache, and a confirmed-events program; retries differ. These digital results do not establish a new memory mechanism or broad service transfer.

The four incomplete cells are L1-G_0, L2-G_0, L2-G_1 and L3-H_0 on the O task. Each final assistant response has stop reason `length`, 32,768 output tokens and 32,768 reasoning tokens, with no final submission. Their actual result files confirm no model-submitted artifact. They are response-token-censored cells, not host cache failures and not submitted service FAIL outcomes. Each remains in the nine-cell arm denominator and in all charged costs; no replacement was run. Dropping them would change the registered comparison.

6.3 Costs of obtaining the result

Full actual G_1 and G_0 pipelines use 138 and 135 requests, with upper estimates USD 1.2447618 and USD 1.2585825. The lower full USD for G_1 does not establish an efficiency advantage: original host incidents cost G_0 twelve requests/USD 0.0742245 and G_1 nine/USD 0.0576228. Removing only those four incidents, while

retaining all recoveries, gives G_1 129 requests/USD 1.1871390 and G_0 123/USD 1.1843580. The direction of the USD difference changes. Common H_0 selection and final controls use 76 requests/USD 0.5154765 (Table 5). This comparison charges learned arms for forming and selecting methods; it does not make H_0 information exposure identical to their longer historical text.

Scope	Full q	Full upper USD	Effective q	Effective upper USD
G_0	135	1.2585825	123	1.1843580
G_1	138	1.2447618	129	1.1871390
H_0 controls	76	0.5154765	76	0.5154765

Table 5: Full arm pipelines include both siblings, proposal/selection/final calls, retained failures, and allocated G_1 formation. Effective pipelines remove only the four original host incidents, retaining all four charged recoveries. H_0 has selection/final controls but no proposal or formation stage.

Final episodes alone cost H_0, G_0, and G_1 45, 40, and 45 requests, respectively, with USD 0.3416751, 0.4800981, and 0.5159961 upper. These are not full pipeline costs. Fewer calls in a censored cell also cannot be read as greater task efficiency. Per-lineage full costs appear in Table 7; neither unselected siblings nor failed, censored, or confirmation calls are omitted.

Across all stages, the campaign closes 77 attempts with 367 uniquely charged provider requests, USD 3.1767228 upper, and 5,951.639484 summed episode wall seconds (Table 6). Full G_0 plus G_1 pipelines, H_0 controls, and separate O0 development conserve these campaign totals. There are 25 new digital scientific responses, zero native dispatches, and zero new UNKNOWN provider-usage records. S0's sealed preservation readback also retains 24 earlier UNKNOWN records; zero new UNKNOWN does not erase them [4]. Exact-input reuse remains distinct from fresh Agent decisions. Codex assistance and historical acquisition are unpriced, so the ledger is not an invoice or complete end-to-end cost.

Stage	Attempts	Requests	Upper USD	Summed wall s
Original-service development	3	18	0.1579020	240.418
G_1 formation	1	3	0.0269157	38.723
Twelve H proposals	12	43	0.4491399	862.946
Selection, including incidents	34	173	1.2049959	2027.954
Final evaluation	27	130	1.3377693	2781.598
Total	77	367	3.1767228	5951.639

Table 6: Conserved campaign accounting. The 34 selection attempts comprise 30 effective cells and four original incidents. Summed episode wall time is not elapsed campaign time. Rates and ledger upper estimates are not invoices.

Original host-incident overhead appears only in lineage 1. Table 7 separates full and effective costs in every arm; the three G_1 shares conserve the one actual formation event.

Lineage arm	PASS	Full q	Full USD	Effective q	Effective USD
L1-G0	2/3	51	0.4392936	39	0.3650691
L1-G1	3/3	53	0.4813594	44	0.4237366
L2-G0	2/3	40	0.4202280	40	0.4202280
L2-G1	2/3	41	0.3678073	41	0.3678073
L3-G0	2/3	44	0.3990609	44	0.3990609
L3-G1	3/3	44	0.3955951	44	0.3955951

Table 7: Per-lineage full and effective costs. Formation is one event with three allocated shares, not three independent formations. Full/effective definitions are identical to Table 5.

7 Limitations and interpretation

The G_0–G_1 comparison holds historical information common, but observes only one formed G_1, three descendant pairs, one provider alias, and a small fixed digital task family. It cannot estimate variation across independently formed improvers or establish broad autonomous improvement. Comparing an informed descendant H with H_0 also changes information embedded in method text. That contrast is not a pure procedural effect. Historical rehearsal can overfit retained observations; task-specific source fields and proposal-role instructions can fail in a new consuming role. The cache incident and explicit recovery must remain part of the interpretation and cost record.

Physical qualification remains UNKNOWN. Earlier fixed-space A/R service `NO_GO`, dark-service `FAIL`, strict FIFO `FAIL`, and product `NOT_ACCEPTED` remain retained evidence boundaries. Digital service outcomes, source readback, structural admission, and a successful cache repair do not establish hardware acceptance. The 8/9 G_1 total is descriptive, conditional on one formed improver and this task sample. No statistical superiority, stable advantage, general transfer capability, or recursive self-improvement is established. The unchanged H_0 remains the default; neither the global method nor the improver is promoted.

8 Reproducibility and conclusion

The binding pins the complete handoff, source results, request ledgers, six selections, frozen methods, and unchanged style/vector logo. The reader recomputes counts, conserved costs, ranking, consumed method composition and table text without a model or scientific rerun. Canonical JSON digests remain distinct from byte hashes; original preparation-error receipts are preserved.

Run `render_phase4.py` with `check` or `build`, using the existing runtime and `-B` from this paper directory. The binding/handoff records absolute commands, fixed TeX dates, two-pass build, fonts, PDF hash/pages and visual QA. S0's single registered final repository check remains pending after import; no SDK, science, init or CI was rerun here. This local report does not assess literature novelty or submission readiness.

The experiment produces one model-authored proposal procedure and twelve actual descendants under controlled information and cost accounting. Selection saturates at PASS, final task differences occur only on O, and four token-censored attempts materially affect completion. The retained apparatus incident changes the full/effective USD comparison. These facts support preserving the default method and the bounded evidence record, rather than declaring an autonomous improvement capability.

Local sealed source records

[1] BF campaign. Phase 4 finite protocol v1.1 and original registration, 20 October 2026. Local records: `protocol-v1.1.json`, `registration.json`. Full byte hashes are in the paper binding.

[2] BF campaign. Closed actual results, final details and descriptive analysis, 20 October 2026. Local records: `actual-results.json`, `final-details.json`, `analysis-summary.json`; bound by `paper-handoff.json`.

[3] BF campaign. Selection completion, selected methods and final selection freeze, 20 October 2026. Local records: `selection-complete.json`, `selected-methods.json`, `final-selection-freeze.json`.

[4] BF campaign. Final source, usage and preservation readback, 20 October 2026. Local record: `preservation-final.json`.

[5] BF campaign. Host incident, cache-trace amendment, qualification and four recovery outcomes, 20 October 2026. Local records: `host-error-hold.json`, `amendment-registration.json`, `cache-repair-qualification.json`, `recovery-results.json`.

[6] BF campaign. Three original-service development transfer episodes, 20 October 2026. Local record: `OO-transfer-results.json`.

A Every final design cell

The task identifiers are:

- Task 1: `p4-bf-stretched-port-window-v1`.
- Task 2: `p4-clite-half-power-query-v1`.
- Task 3: `p4-o-unexposed-waveform-v1`.

L identifies a descendant lineage slot, conditional on the same shared G_1 formation. All registered cells, including non-submissions, are shown. Parameters are rounded only for display; the bound JSON retains exact submitted values.

Task	L	Arm	Result	Design parameters	q	Upper USD
1	1	H_0	P	10 / 0.005	5	0.0221046
1	1	G_0	P	14 / 0	4	0.0378396
1	1	G_1	P	10 / 0	5	0.0352167
1	2	H_0	P	22 / 0	5	0.0289770
1	2	G_0	P	22 / 0	6	0.0786750
1	2	G_1	P	10 / 0	5	0.0256398
1	3	H_0	P	22 / 0.05625	5	0.0335634
1	3	G_0	P	22 / 0.05625	5	0.0416196
1	3	G_1	P	15 / 0.05625	5	0.0593052
2	1	H_0	P	40 / 6.5	5	0.0218922
2	1	G_0	P	40 / 7.5	5	0.0543483
2	1	G_1	P	50 / 6.5	5	0.0664434
2	2	H_0	P	50 / 6.5	5	0.0242127
2	2	G_0	P	50 / 6.89	5	0.0509238
2	2	G_1	P	50 / 6	5	0.0445269
2	3	H_0	P	50 / 6.5	5	0.0268155
2	3	G_0	P	50 / 7	4	0.0327336
2	3	G_1	P	50 / 6.5	5	0.0415404
3	1	H_0	F	1 / 1	5	0.0606894
3	1	G_0	C	—	3	0.0549690
3	1	G_1	P	0.75 / 0.25	6	0.0990039
3	2	H_0	P	0.75 / 5	7	0.0753684
3	2	G_0	C	—	3	0.0501078
3	2	G_1	C	—	3	0.0581409
3	3	H_0	C	—	3	0.0480519
3	3	G_0	F	1 / 0.25	5	0.0788814
3	3	G_1	P	0.75 / 1	6	0.0861789

Table 8: Every final cell. P: submitted PASS; F: submitted FAIL; C: token-censored, no submission. Tasks 1/2 report capacitance (μF) / transmit gate (s); task 3 reports sample period (s) / retry (s). All submitted task-3 designs use 15 mF, nonvolatile cache and confirmed-events; other fields and complete JSON artifacts remain bound in final-details. Disagreements are task 3, L1-H_0 and L3-G_0, both judged PASS by the participant but FAIL by the host.